Exhibits for the paper (preview)

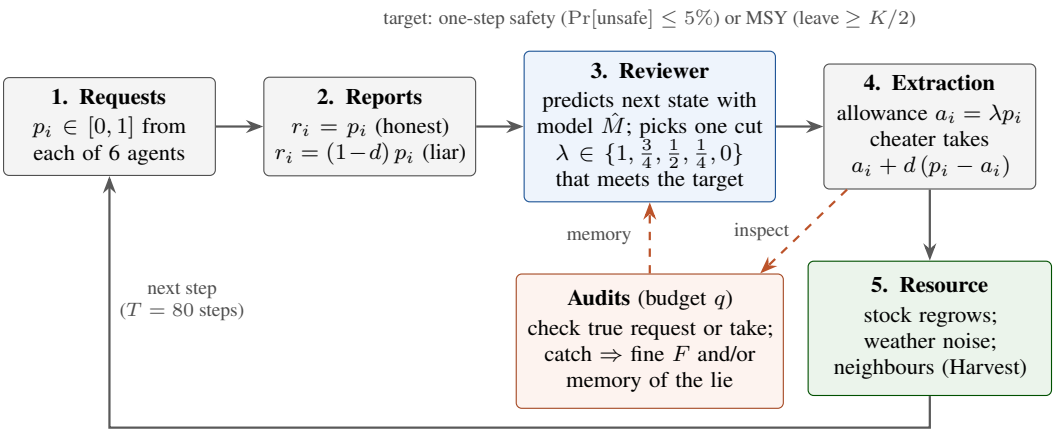


Figure 1: One step of the audited commons. Six agents request shares of a renewable resource; a reviewer with a model of the resource chooses one shared cut that meets its target; agents extract, and some may take more than allowed. With budget q, audits check a few agents and turn what they find into consequences (a fine, a memory of the lie, or both).

Table 1: The three games. All share the step in Figure 1; they differ in how agents' actions combine into harm.

	Fishery	Harvest	Synergistic Pollution
Resource	one stock, $K = 100$	6 patches on a ring, max 20 each	river quality, max 100
Dynamics	logistic, $r = 0.7$	logistic, $r = 0.476$, weather $\sigma = 0.42$	logistic, $r = 1$, weather $\sigma = 3$
Coupling between agents	shared stock (additive)	over-harvest spills to neighbours (0.22)	damage $\kappa XY + \lambda(X + Y)$ (compositional)
Agents	6 (2 or 4 stress)	6 (2 or 4 stress)	6 (3 per reagent)
Max request per agent	6 per step	6 per step	1 per step
One-step target	stock ≥ 10 next step	$\Pr[\text{mean health} < 10] \leq 5\%$	$\Pr[Q' < 30] \leq 5\%$
MSY target	leave $\geq K/2$ after harvest	—	—
Can cheating pay?	yes	no (harm falls on own patch)	not tested
Horizon, test contexts	80 steps, 64	80 steps, 64	80 steps, 64
Used in	R1, S1–S5, R2, L1	R1, S1, S1b, S3, R2	C1

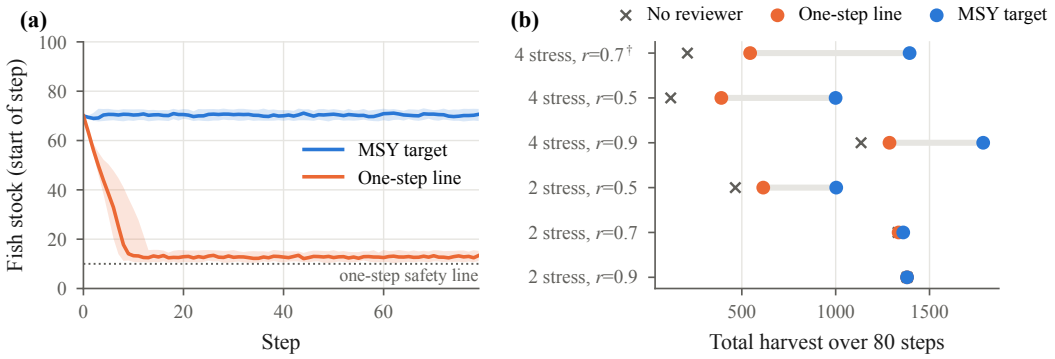


Figure 2: The goal matters more than inspection. (a) Fishery stock under the joint reviewer inspecting all six requests (4 stress agents, $r = 0.7$; line = median, band = 10th–90th percentile over 64 contexts). Under the one-step line the reviewer approves running the stock down to just above 10; the MSY target holds it near 70. (b) Total harvest over 80 steps in the six Fishery settings of the predeclared grid († piloted setting). MSY gives more harvest wherever the one-step line binds (e.g. 1,394 vs 544 in the piloted setting) and ties where it does not.

Table 2: Audit regimes compared (Fishery, MSY target, cheaters choose one shared over-take level; 64 held-out contexts). Gain: cheaters' net gain over complying, group of four, 80 steps. Honest harvest: per honest agent. Checks per step include the extra checks that "memory + cap" spends on known cheaters. Bold: best in column.

Regime	q	Cheaters' gain [95% CI]	Honest harvest	Steps breaking MSY (%)	Checks per step
No audits	—	112.8 [106, 120]	39.5	100	0.00
Fine	0.10	**0.0**	**104.1**	**0**	0.61
	1/6	**0.0**	**104.1**	**0**	1.00
Memory	0.10	58.4 [49, 68]	74.5	54	0.61
	1/6	55.1 [46, 65]	77.1	49	1.00
Memory + cap	0.10	−3.5 [−36, 14]	95.2	16	3.78
	1/6	8.3 [7, 10]	99.4	10	4.07
Memory + shared cut	0.10	141.7 [136, 147]	36.0	24	0.61
	1/6	142.3 [137, 148]	36.0	16	1.00
Fine + memory + cut	0.10	**0.0**	**104.1**	**0**	0.61
	1/6	**0.0**	**104.1**	**0**	1.00
Everyone complies	—	0	104.1	0	0

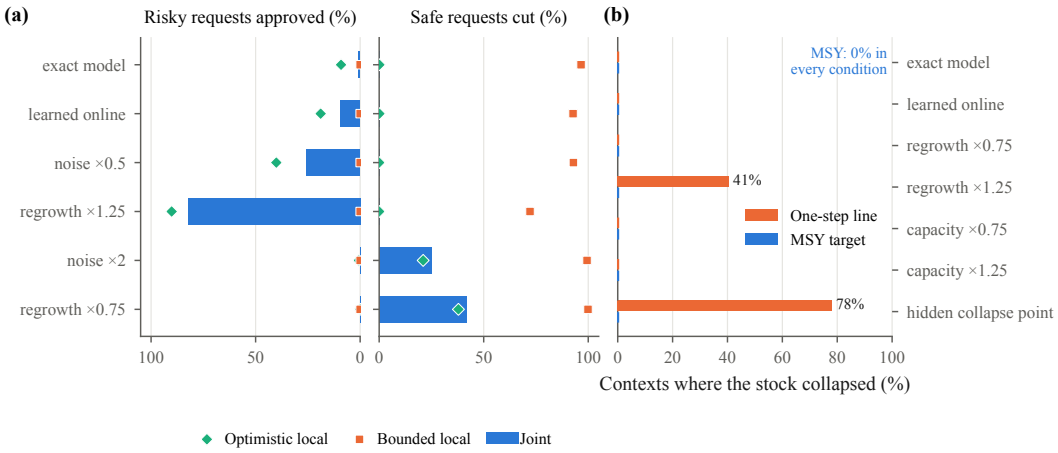


Figure 3: A wrong model breaks an otherwise exact reviewer; the MSY target protects against a wrong model form. (a) Harvest (piloted setting), decisions on 2,734 resolved requests when the reviewer's model is wrong in one parameter. The direction of the error decides which mistake is made: underestimating noise or overestimating regrowth lets risk through (joint: 0.9% $\rightarrow$ 25.6% and 82.3%); the opposite errors cut safe requests (25% and 42%). A reviewer that learns the model online stays close to exact. (b) Fishery (4 stress agents, $r = 0.7$), closed loop over 64 contexts: with the one-step line, the stock collapses in 41% of contexts when regrowth is overestimated and in 78% when the true dynamics have a hidden collapse point (Allee threshold 20); with the MSY target it never collapses.

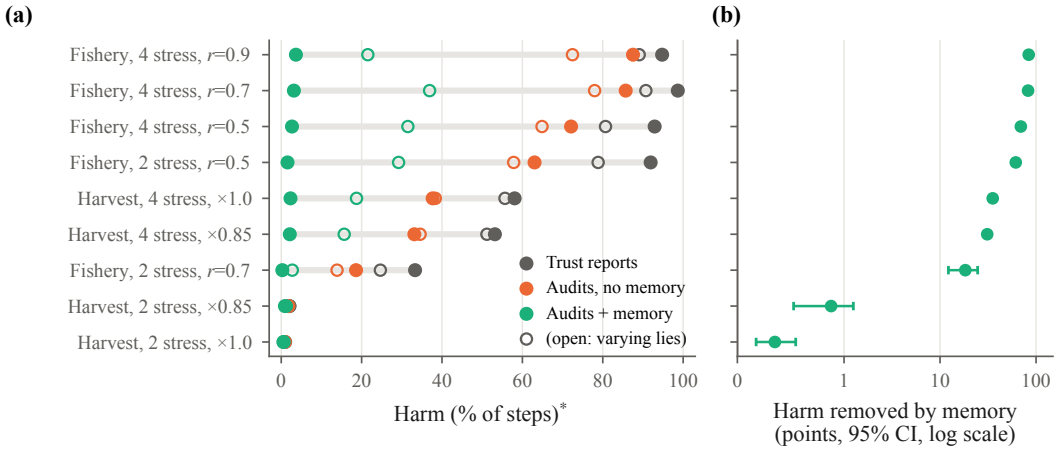


Figure 4: Audits help through memory. (a) Harm in each of the nine testable settings of the predeclared grid when two or four agents under-report and the reviewer trusts reports, audits without memory, or audits and remembers each caught lie (one random audit per step, no sanctions). Filled: lies of a fixed size; open: lies of a random size each step. *Fishery: steps breaking the MSY target; Harvest: steps in an unsafe state. (b) Harm removed by memory relative to audits without memory, fixed-size lies, with 95% paired context-bootstrap intervals; every interval excludes zero.

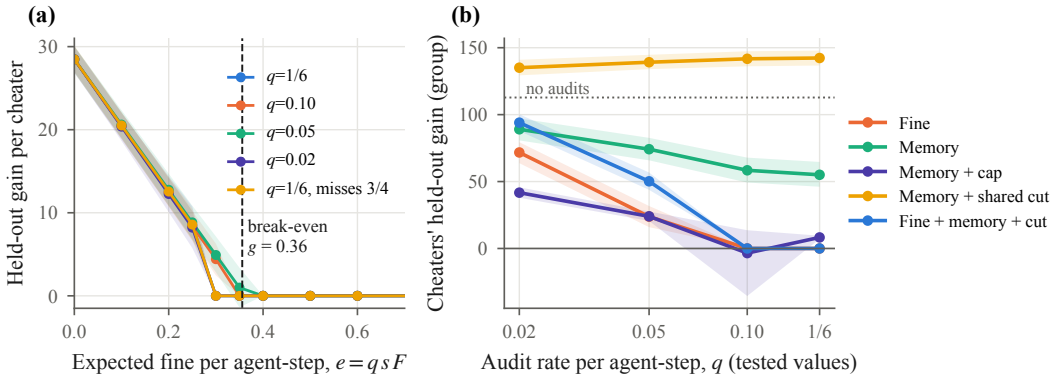


Figure 5: Fines deter at the expected-fine break-even; memory alone does not deter. (a) Fishery: held-out gain per cheater (80 steps) at the cheating level chosen on training contexts, against the expected fine per agent-step $e = q\,s\,F$, for four audit rates and for audits that miss three of four over-takes. Curves overlap because only e matters; every group stops between $e = 0.30$ and 0.40, around the break-even $g = 0.36$ (the no-check gain per over-take step). (b) Cheaters' held-out gain against audit rate for five audit regimes (fine $F = 6$). A fine deters from $q = 0.10$; memory alone keeps about half the gain; tightening the shared cut for known cheaters raises their gain above no audits. Bands: 95% paired context-bootstrap intervals; 64 held-out contexts.

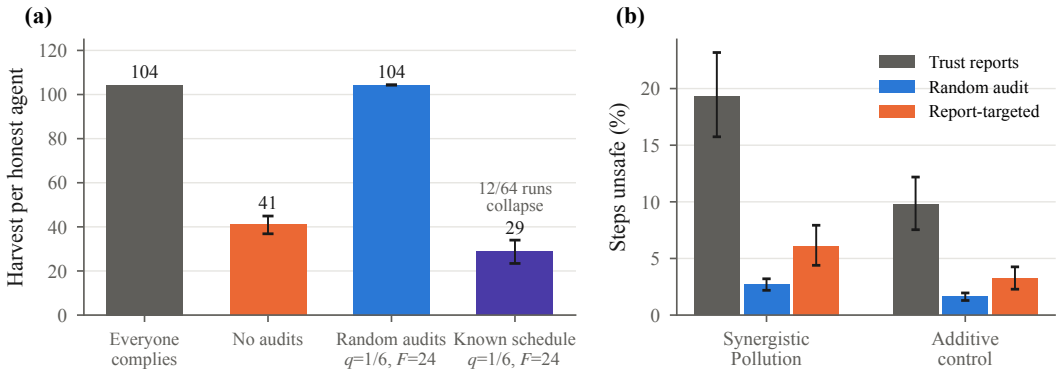


Figure 6: How audits are aimed matters. (a) Fishery: harvest per honest agent with the same audit rate ($q = 1/6$) and fine ($F = 24$). Random audits deter completely; a known schedule (every agent on every sixth step) lets cheaters take everything on unaudited steps, leaving honest agents worse off than with no audits (28.8 vs 40.9). (b) Synergistic Pollution and its additive control: with two under-reporters and one audit per step with memory, aiming the audit at the most influential report gives more unsafe steps than auditing at random (+3.4 points [1.8, 5.3] in the compositional game). This was a post hoc finding; experiment T1 tests it in all three games. Error bars: 95% paired context-bootstrap intervals.